CAUTION

KEEP OUT OF REACH OF CHILDREN
READ SAFETY DIRECTIONS BEFORE OPENING OR USING

PENNCOZEB®
FUNGICIDE

ACTIVE CONSTITUENT: 800 g/kg MANCOZEB

GROUP Y FUNGICIDE

For the control of certain fungus diseases of field crops, vegetables, fruit, tobacco, turf and ornamentals as per the DIRECTIONS FOR USE table

READ THE DIRECTIONS FOR USE THOROUGHLY BEFORE USING THIS PRODUCT

Contents: 25 kg

N.R.A.
APPROVED LABEL (TEXT)
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Date: 8/05/2002
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Directions For Use:

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
FIELD CROPS Peanuts	Cercospora leaf Spot	QLD, NSW, and WA only		1.7 kg	14 Days	Apply at 10 to 14 day intervals commencing when disease symptoms first appear. Use the higher rate and shorter intervals during weather conditions favourable to disease development. Add surfactant such as Spraymate Bond or Chemwet 1000 at 40 ml/ha.
	Rust	QLD & WA				
Soybeans	Rust	All States		2.2 kg	7 days	Spray in 200-400 L water per hectare for ground equipment and 50 L water for aerial application. Commence spraying at first sign of disease and then at 7-10 day intervals while warm, rainy weather conditions prevail. Add a surfactant such as Chemwet 1000.
Tobacco	Blue mould Brown spot	NSW, ACT, QLD, Vic & WA only	200 g	2.2 kg	7 days	Spray at 7day intervals following a spray at transplanting. **Seedbed treatments:** Spray to wet seedlings without runoff. Start spraying when plants are 1.5cm across then twice weekly. **Field treatments:** Apply at 7 day intervals commencing 10 days after transplanting and ceasing 7 – 10 days before the first harvest. Increase the spray volume accordingly as plant size increases.
	Frog eye leaf spot	Qld & WA only				
Poppies	Downy mildew	Tas only	-	2.5 kg	7 weeks	Apply as a protectant spray when weather conditions favour disease development. Repeat spray on 14 day intervals if conditions persist.

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
VEGETABLES Beans	Rust Anthracnose	All States	150-200 g	1.7-2.2 kg	7 days	Spray at first sign of disease. Repeat at 7-10 day intervals. For improved coverage and spray retention add Spraymate Bond or Chemwet 1000 adjuvant
	Angular leaf spot	Qld, WA, NSW & ACT only				
Broadbeans (including faba beans)	Chocolate spot (*Botrytis fabae*)	NSW, ACT, Tas, Vic, WA & SA only				
	Suppression of leaf blight (*Ascochyta fabae*)	NSW, ACT, Tas, WA & SA only		1.7-2.5 kg		
	Rust (*Uromyces viciae-fabae*)	Tas, SA, WA, Vic, NSW & ACT		1.7-2.2 kg		
Carrots	Cercospora leaf spot Alternaria leaf spot	All States	150-200 g	1.7-2.2 kg	7 days	Spray at first sign of disease, then at 7-10 day intervals or more frequently depending on blight conditions.
Celery	Septoria leaf spot (leaf blight)	All States	150-200 g	1.7-2.2 kg	7 days	Spray at first sign of disease. Repeat at 7-10 day intervals.
COLE CROPS Cabbage Cauliflower Brussel sprouts	Anthracnose Downy Mildew Alternaria spot Ring spot	All States	150-200 g	1.7-2.2 kg	7 days	Spray at first sign of disease. Repeat at 7-10 day intervals. Add a surfactant such as Chemwet 1000.
Broccoli		All States except Vic				
CUCURBITS Cucumber Canteloupe Pumpkin Melons Zucchini Squash	Downy Mildew Anthracnose Gummy stem blight	All States	150-200 g	1.7-2.2 kg	7 days	Apply when disease symptoms first appear and then repeat at 7-10 day intervals. Use higher rate and shorter intervals during weather conditions favourable to disease development. For improved coverage and spray retention add a surfactant such as Chemwet 1000.
Pumpkin	Septoria spot	Qld & WA only				
Capsicum	Target spot	All States	150-200 g	2 to 3 kg	14 days	Spray at first sign of disease. Repeat at 7-10 day intervals. Use higher rate and shorter intervals during weather conditions favourable to disease development.

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
Lettuce	Downy Mildew Anthracnose Septoria leaf blight or spot	All States	150-200 g	1.7-2.2 kg	14 days	Apply when disease symptoms first appear and then repeat at 7-10 day intervals. Use higher rate and shorter intervals during weather conditions favourable to disease development.
Onion and Garlic	Downy mildew Purple blotch	All States	150-200 g	2.2-3.5 kg	7 days	
Potatoes	Early blight (Target spot) Late blight (Irish blight)	All States	150-200 g	1.7-2.2 kg	7 days	Apply when disease symptoms first appear and then repeat at 7-10 day intervals. Use higher rate and shorter intervals during weather conditions favourable to disease development.
Potato seed pieces	Fusarium seed piece decay	WA only	250 g		—	Dip whole or cut potato tubers. Place treated pieces in clean containers and plant as soon as possible. Spread in cool place if held before planting. Do not use treated seed potatoes for human consumption or stock feed.
Rhubarb	Downy mildew	All States	150-200 g	1.7 – 2.2 kg	14 days	Apply when disease symptoms first appear and then repeat at 7-10 day intervals. Use higher rate and shorter intervals during weather conditions favourable to disease development.
	Rust	NSW, ACT, WA & Tas only				
Spinach Silverbeet Beetroot	Downy mildew Cercospora leaf spot	All States	150-200 g	1.7 – 2.2 kg	14 days	Add a surfactant such as Chemwet 1000.

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
Tomatoes	Early blight (Target spot) Late blight (Irish blight)	All States	150-200 g		14 days	Spray at first sign of disease. Repeat at 7-10 day intervals. Do not apply to tomato seedlings before transplanting. Use higher rate and shorter intervals during weather conditions favourable to disease development.
	Anthracnose Leaf mould (*Fulvia fulva*)					
	Grey leaf spot Phoma rot	NSW, ACT, WA & QLD only				
POME FRUITS Apples Pears	Apple Scab (Black spot)	All States	**Dilute Spraying** 150-200 g **Concentrate Spraying** Refer to the Application section		14 days	Spray at 7-14 day intervals following green tip copper spray. Use higher rate and shorter spray interval during prolonged wet conditions. Product not suitable for use in integrated mite control programs (Tas. only). Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
	Target spot Bitter rot Sooty blotch Fly speck	All States				
	Ripe fruit spot	QLD & WA only				
STONE FRUITS Peaches Apricots Plums (except Wilson plums)	Rust Shot hole Freckle	All States	**Dilute Spraying** 150-200 g **Concentrate Spraying** Refer to the Application section		14 days	Spray at early bloom (1-10%) mid to full bloom (50-100%) petal and shuck fall, then at intervals of 14 days. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
	Brown rot	Vic, Tas, WA, SA, NSW & ACT only				
Cherries		Vic, Tas, & WA only				

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
STONE FRUITS Nectarines	Brown rot	SA, WA & Tas only	**Dilute Spraying** 150-200 g **Concentrate Spraying** Refer to the Application section		14 days	Spray at early bloom (1-10%) mid to full bloom (50-100%) petal and shuck fall, then at intervals of 14 days. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
Almonds		SA & WA only				
OTHER FRUITS Grapevines	Black spot	All States	**Dilute Spraying** 200 g **Concentrate Spraying** Refer to the Application section		30 days	For black spot control spray commencing at bud burst. Repeat every 14 days. If downy mildew is expected begin spray application at 10-15 cm stage of shoot growth. Continue the program at intervals of 10-14 days until disease threat is over. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
	Downy mildew	All States				
	Dead arm	Qld & WA only	**Dilute Spraying** 150-200 g **Concentrate Spraying** Refer to the Application section		30 days	Spray at bud burst and repeat 7-10 days later. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
	Phomopsis Cane and Leaf spot	All States			30 days	Apply at budburst and then repeat 7 to 10 days later. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product to the target crop whether applying this product by dilute or concentrate spraying methods

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
Mangoes	Anthracnose *Colletotrichum gloeosporiodes*	Qld, NSW & WA only	**Dilute Spraying** 200 g **Concentrate Spraying** Refer to the Application section		14 days	Spray at weekly intervals during flowering then monthly until harvest. Use sufficient water for good coverage. Add surfactant such as Chemwet 1000. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
Passionfruit	Septoria spot Brown spot (*Alternaria passiflorae*)	Qld & WA only	**Dilute Spraying** 150 g **Concentrate Spraying** Refer to the Application section		14 days	Apply at 10-14 day intervals from October to May and every 21-28 days for remainder of year. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
	Anthracnose	NSW, ACT & WA only	**Dilute Spraying** 200 g plus 600 ml miscible Summer oil **Concentrate Spraying** Refer to the Application section			Spray at weekly intervals during flowering then fortnightly near harvest. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
Citrus	Black spot	NSW, ACT, Tas, Vic and WA only QLD only	**Dilute Spraying** 200 g plus 250 ml white oil **Concentrate Spraying** Refer to the Application section **Dilute Spraying** 200 g plus 600 ml white oil **Concentrate Spraying** Refer to the Application section		—	**Heavy Soil:** Spray at 8 and 15 weeks following a Bordeaux Spray (5-5-100) at petal fall. **Light Soil:** Spray at 6 and 13 weeks following a Bordeaux Spray (3.5-3.5-100) plus 100-140 mL Spraymate Bond Adjuvant per 100L of spray. In Qld. use a Bordeaux Spray mixture of 8-8-100 at petal fall. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
	Citrus rust mite Brown citrus mite	QLD & WA only	**Dilute Spraying** 200 g **Concentrate Spraying** Refer to the Application section		—	Spray at first sign of rust or citrus mite (usually December to May) Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.
	Citrus bud mite	QLD & WA only			—	Bud mite: Apply when blossom malformation is observed (usually February). Apply by dilute or concentrate spraying equipment. Apply the same total amount of product whether applying this product by dilute or concentrate spraying methods.

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
Bananas	Cercospora leaf spot (Sigatoka Disease)	NSW & WA only	200-400 g	2.2 kg) + 3.5-4.0 L miscible summer oil	14 days	Apply when weather conditions favour disease outbreak. Repeat treatments in accordance with locally recommended spray schedules. Use sufficient water to ensure coverage. Apply at three week intervals from December / January until May. **North Queensland:** Apply every two weeks during the wet season. Extend to three weeks after this and then four to five weeks during the drier months returning to three weeks prior to the next season. Apply at 3 weeks intervals from December or January to May. NORTH QLD: Apply every two weeks during the wet season. Extend to 3 weeks after this and then to 4-5 weeks during the drier months returning to 3 weeks prior to the next wet season.
	Leaf spot Leaf speckle Cordana leaf spot Fruit freckle and Black pit	QLD & WA only	200 g plus 600 ml summer oil	2.2-4.5 kg plus 3.5-4.2 L summer oil or water miscible oil		
		Nth QLD only	200 g	2.2 kg		

CROP	DISEASE	STATE	RATE/100L	RATE/ HECTARE	WITH-HOLDING PERIOD	CRITICAL COMMENTS
ORNAMENTALS Carnations	Rust Alternaria leaf spot	Tas & WA only	150-200 g		_	Apply when disease symptoms first appear and then repeat at 7-14 day intervals.
Flowers, Shrubs	Septoria leaf spot	NSW, ACT & WA only	150-200 g		_	Apply when disease symptoms first appear and then repeat at 7-10 day intervals.
Ferns	Botrytis leaf spot		75-100 g		_	
Gladioli Chrysanthemums	Grey mould (*Botrytis*) Septoria leaf spot	All States	150-200 g		_	
	Rust	Tas & WA only				
Roses	Black spot	All States	150-200 g		_	Apply when disease symptoms first appear and then repeat at 7-10 day intervals.
Turf	Fusarium Rhizoctonia Helmintho-sporium	All States	_	200-250 g/ 100 sq. metres	_	Spray in sufficient water for adequate coverage. Apply every 7 days throughout susceptible periods.

NOT TO BE USED FOR ANY PURPOSE, OR IN ANY MANNER, CONTRARY TO THIS LABEL UNLESS AUTHORISED UNDER APPROPRIATE LEGISLATION

WITHHOLDING PERIOD

BEANS, CARROTS, CELERY, COLE CROPS, CUCURBITS, GARLIC, ONION, POTATOES, SOYBEANS, TOBACCO – DO NOT APPLY LATER THAN 7 DAYS BEFORE HARVEST.

ALMONDS, BANANAS, BEETROOT, CAPSICUM, LETTUCE, MANGOES, NECTARINES, PASSIONFRUIT, PEANUTS POME FRUIT, RHUBARB, SILVERBEET, SPINACH, STONE FRUIT, TOMATOES – DO NOT APPLY LATER THAN 14 DAYS BEFORE HARVEST.

GRAPEVINES – DO NOT APPLY LATER THAN 30 DAYS BEFORE HARVEST. POPPIES – DO NOT APPLY LATER THAN 49 DAYS BEFORE HARVEST. DO NOT GRAZE LIVESTOCK ON CROPS INCLUDING PEANUTS AND SOYBEANS TREATED WITH THIS PRODUCT OR FEED TREATED CROPS AND/OR TRASH TO LIVESTOCK. DO NOT FEED GRASS CLIPPINGS FROM TREATED TURF AREAS TO POULTRY OR ANIMALS.

GENERAL INSTRUCTIONS:

Before opening, carefully read Directions For Use, Precautionary Statements, Safety Directions and First Aid instructions. This product is a protectant fungicide for the control of certain fungus diseases in a number of crops. For best results, apply product thoroughly to the treated crop in a regular spray program.

FUNGICIDE RESISTANCE WARNING

GROUP Y FUNGICIDE

Penncozeb® Fungicide is a member of the Multi-site activity group of fungicides. For fungicide resistance management Penncozeb® Fungicide is a Group Y fungicide.

Some naturally occurring individual fungi resistant Penncozeb® Fungicide and other Group Y fungicides may exist through normal genetic variability in any fungal population. The resistant individuals can eventually dominate the fungi population it these fungicides are used repeatedly. These resistant fungi will not be controlled by Penncozeb® Fungicide and other Group Y fungicides, thus resulting in a reduction in efficacy and possible yield loss.

Since the occurrence of resistant fungi is difficult to detect prior to use, ATOFINA (Australia) Pty Limited accepts no liability for the losses that may result from the failure of Penncozeb® Fungicide to control resistant fungi.

MIXING

Partly fill spray tank with water and agitate while adding this product slowly. Continue agitation after filling and during spraying to ensure spray mixture is uniform and to prevent product settling out. For low volume or aerial application prepare a slurry or premix when adding to the spray tank. Additional insecticides, fungicides, surfactants, etc. should be added after the product is in suspension. Rinse out spray tank pumps and nozzles at the end of the day.

ADJUVANTS

The use of Spraymate Bond Adjuvant may improve spray deposition and rainfastness and should be used according to label directions.

APPLICATION:

High Volume Dilute Spray. This product may be applied as a high volume dilute spray to fruit tree and vine crops and some row crops. Apply sufficient spray volume to wet all leaf surfaces to the point of runoff. The spray equipment should be adjusted so that the spray is evenly distributed, thoroughly covering all plant surfaces.

Dilute Spraying

- Use a sprayer designed to apply high volumes of water up to the point of run-off and matched to the crop being sprayed.
- Set up and operate the sprayer to achieve even coverage throughout the crop canopy. Apply sufficient water to cover the point to the point of run-off. Avoid excessive run-off.
- The required water volume may be determined by applying different test volumes, using different settings on the sprayer, from industry guidelines or expert advice.
- Add the amount of product specified in the Directions for Use table for each 100L of water. Spray to the point of run-off.
- The required dilute spray volume will change and the sprayer set up and operation may also need to be changed as the crop grows.

The spray volume will vary according to the treated crop and should be increased when spraying large or dense plants to ensure thorough coverage. The spray volume should also be increased throughout the growing season as the crop increases in size. Suggested water volumes for mature tree/vine crops are as follows:
Vine crops - 200 to 1100 L/ha.
Pome and Stone Fruit - 1100 to 4000 L/ha.
Citrus - 4000 to 8000 L/ha.
These volumes are a guide only and appropriate volumes should be determined by the methods outlined above.

Concentrate Spraying

- Use a sprayer designed and set up for concentrate spraying (that is a sprayer which applied water volumes less than those required to reach the point of run-off) and matched to the crop being sprayed.
- Set up and operate the sprayer to achieve even coverage throughout the crop canopy using your chosen water volume.
- Determine an appropriate dilute spray volume (See Dilute Spraying above) for the crop canopy. This is needed to calculate the concentrate mixing rate.
- The mixing rate for concentrate spraying can be then calculated in the following way:

  EXAMPLE ONLY

  1. Dilute spray volume as determined above: For example 1000 L/ha
  2. Your chosen concentrate spray volume: For example 250 L/ha
  3. The concentration factor in this example is: 4 X (i.e. 1000 L/250 L = 4)
  4. If the dilute label rate is 250 mL/100 L, then the concentrate rate becomes 4 x 250, that is 1 L/100 L of concentrate spray.

- The chosen spray volume, amount of product per 100 L of water, and the sprayer set up and operation may need to be changed as the crop grows.

- For further information on concentrate spraying, users are advised to consult relevant industry guidelines, undertake competency training and follow industry Best Practices.

Concentrate and ground application sprays

This product may be applied as a concentrate spray through low volume air blast equipment or as a ground application through spray booms. Adjust the rate to ensure that the correct amount is applied per hectare from the particular equipment used. Use the higher rate on larger plants and when weather conditions favour the development of disease.

Aerial application

This product may be applied by agricultural aircraft. Use at least 30 to 50 litres of spray mixture per hectare. Reference should be made to the Directions for Use table for the correct rate of application in specific crop situations.

SPRAY TIMING

Apply treatments when weather conditions favour disease outbreak and continue until the threat of disease has passed. Repeated applications are necessary to protect new plant growth and as conditions favour disease development, reduce the spray interval.

COMPATIBILITY

This product may be harmful to *Typholodromus pyri* and should not be used in orchards where integrated control methods are practiced. Most commonly used agricultural fungicides and insecticides are compatible with Penncozeb® Fungicide, however reference should be made to the other manufacturer's recommendations before preparing tank mixes.

PROTECTION OF WILDLIFE, FISH, CRUSTACEA AND ENVIRONMENT

Dangerous to fish.
DO NOT contaminate streams, rivers or waterways with the chemical or used containers.
DO NOT feed crops or trash to livestock.

STORAGE AND DISPOSAL

Store in the closed, original container, in a well ventilated area away from children, animals, food, feedstuffs, seed and fertilizers. Do not store for prolonged periods in direct sunlight. Do not allow container to become wet or over-heated in storage.
Users are reminded that this product should be used before the stated expiry date. Single rinse before disposal. Add rinsings to spray tank. Do not dispose of undiluted chemical on site. Puncture or shred and bury empty containers in a local authority landfill. If there is no local authority landfill readily available in your area, bury the containers below 500mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desired vegetation and tree roots. Empty containers and product should not be burnt.

SAFETY DIRECTIONS

Harmful if inhaled or swallowed. Will irritate the eyes, nose, throat and skin. Avoid contact with eyes and skin. Do not inhale dust or spray mist. When preparing the spray wear cotton overalls buttoned to the neck and wrist, a washable hat, elbow-length PVC gloves and face shield or goggles. After use and before eating, drinking or smoking wash hands, arms and face thoroughly with soap and water. After each day's use wash gloves, face shield or goggles and contaminated clothing.

FIRST AID

If poisoning occurs, contact a Doctor or Poisons Information Centre – Phone 131126. Avoid giving alcohol.

Emergency Advice

In case of emergency: Phone 1800 033 498 and ask for shift supervisor

MATERIAL SAFETY DATA SHEET

Additional information is listed in the Material Safety Data Sheet that can be obtained from the supplier.

CONDITIONS OF SALE

"Nufarm Limited ('Nufarm') shall not be liable for any loss, injury, damage or death whether consequential or otherwise whatsoever or howsoever arising whether through negligence or otherwise in connection with the sale, supply, use or application of this product. The supply of this product is on the express condition that the purchaser does not rely on Nufarm's skill or judgment in purchasing or using the same and very person dealing with this product does so at his own risk absolutely. No representative of Nufarm has any authority to add to or alter these conditions ".

NRA Approval No: 53986/0402.
Batch No.:
Date of Manufacture:
Expiry Date: 2 years after production date

BARCODE